



Birla Institute of Technology & Science, Pilani

Pilani Campus

ACADEMIC— UNDER GRADUATE STUDIES DIVISION

First Semester 2025-26

Course Handout Part II

Date: 1st August 2026

In addition to part I (General Handout for all courses appended to the Timetable), this portion gives further specific details regarding the course.

Course Number : BITS F219

Course Title : PROCESS ENGINEERING

Instructor-in-charge : ANIL JINDAL

1. **Course description:** Basic concepts related to heat transfer, mass transfer and flow of fluid, processes and equipment involved in extraction and filtration; mixing and granulation; size reduction and classification, evaporation and distillation, drying and crystallization, humidification and dehumidification. Materials of construction. Theory of compression and consolidation of solids.
2. **Scope and Objective:** This course introduces students to the fundamental principles and practical aspects of pharmaceutical engineering, with a focus on key unit operations involved in the development and manufacturing of pharmaceutical formulations. It covers a wide range of processes including size reduction and separation, drying, evaporation, distillation, mixing, filtration, centrifugation, extraction, compression, crystallization, and materials of plant construction. The course emphasizes the selection, operation, and application of various equipment used in the pharmaceutical industry.
3. **Course learning outcomes:**
 - a. Complete understanding of the principles of key pharmaceutical unit operations
 - b. Knowledge of the construction, working, and applications of various equipment used in pharmaceutical unit operations such as size reduction, drying, distillation, mixing, filtration, centrifugation, and extraction.
 - c. Ability to apply principles of pharmaceutical engineering to analyze and optimize unit operations for enhanced process efficiency, cost-effectiveness, and product quality in pharmaceutical manufacturing.

4. Textbooks and references:

TB1. Lachman, L., Lieberman, H. A., Kanig, J. L., The Theory and Practice of Industrial Pharmacy, Varghese Publishing House, Bombay, 3rd Edition, 1987.

TB2. Aulton M.E., Pharmaceutics: The Science of Dosage form Design, ELBS, 1st Ed.

RB1. Carter, S.J. (Ed.), Cooper and Gunn's Tutorial Pharmacy, Kothari, Bombay, 6th Edition, 1972.

5. Course Plan: weekly schedule for 14 weeks with [3] lectures per week

Lectures	Lecture session/Tutorial Session	Reference	Learning objectives
1	Introduction to process engineering	[Class notes]	To understand the importance of different unit operations in pharmaceutical formulation development
2-4	Basics of flow of fluids, heat transfer and mass transfer	[RB1 §§ 11-13]	To understand basic principle involved in the flow of fluids, heat transfer and mass transfer
5-7	Size Reduction	[TB2 §10]	To understand the principle, construction and working of different mills used in the pharmaceutical industry
8-9	Size Separation	[TB2 §10]	To study the different method of size separation including sieving, microscopy, sedimentation and coulter counter, To understand the criteria of selection of suitable separation method
10-11	Particle size analysis	[TB2 §9]	To determine the particle size of the powder blend, presentation and interpretation of particle size data.
12-14	Evaporation	[RB1 §10]	To understand principles, construction, working, uses, merits and demerits of Steam jacketed kettle, horizontal tube evaporator, climbing film evaporator, forced circulation evaporator, multiple effect evaporator& Economy of multiple effect evaporator.
15-17	Distillation	[RB1 §23]	To study basic principles and methodology of simple distillation, flash distillation, fractional distillation, distillation under reduced pressure, steam distillation & molecular distillation
18-20	Drying	[TB1 §3]	To understand mechanism of drying and to study principles , construction, working, uses, merits and demerits of Tray dryer, drum dryer spray dryer, fluidized bed dryer, vacuum dryer, freeze dryer.



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Lectures	Lecture session/Tutorial Session	Reference	Learning objectives
21-23	Mixing; Liquid, Solid and Semisolid	[TB1 §1]	To study Principles, Construction, Working, uses, Merits and Demerits of Double cone blender, twin shell blender, ribbon blender, Sigma blade mixer, planetary mixers, Propellers, Turbines, Paddles & Silverson Emulsifier,
24-26	Filtration	[TB1 §7]	To study mechanism of filtration, to understand the working principle of various industrial filters
27-29	Centrifugation	[RB1 §21]	To understand principles, construction, working, uses, merits and demerits of Perforated basket centrifuge, Non-perforated basket centrifuge, semi continuous centrifuge & super centrifuge.
30-32	Extraction	[RB1 §22]	To understand the principle and working of different extractors used in the pharmaceutical industry
33-35	Compression and consolidation	[TB1 §4]	To understand the behavior of solids under high pressure during the compression
36-38	Crystallization	[Class notes]	To study principle and working of equipments used for the crystallization in pharmaceutical and chemical processes
39-42	Materials of Pharmaceutical Plant Construction, Corrosion and its prevention	[RB1 §25]	To understand different factors affecting selection of material of construction, identification of suitable material for construction of pharmaceutical equipment/plant

6. Evaluation Scheme:

(Given below are suggestions for possible kinds of evaluation components. Indicative only.)

Component	Weightage	Duration	Date and time	Comments
[Continuous Evaluation]	[10%]	[30 min]	TBA	[Best 3 of 4, Closed book]

Lab component	[20%]	In lab		
[Mid Semester]	[30%]	90 min	As announced in the timetable	[Open/Closed book]
Comprehensive	[40%]	180 min	As announced in the timetable	[Closed book]

Criterion for NC: NC report may be awarded when a student score is below 30% of the average of the top three performers or 40% of median, whichever is lower.

7. **Attendance Policy:** Each student is expected to attend all classes and participate in all evaluation components.
8. **Chamber consultation hours:** As announced in the class.
9. **Makeup Policy:** Makeup will be given only for genuine cases (accompanied by proof) for Mid-sem or comprehensive exam only.

[signature]

Instructor-in-charge

[Course Number]